The Unit Circle and Exact Values

ANSWER KEY

Section 1: Coordinates on the Unit Circle

1 On the unit circle, what do the coordinates (x, y) of a point at angle θ represent?

a. $x = \cos\theta$ **and** $y = \sin\theta$ ✓

b. $x = \sin\theta$ and $y = \cos\theta$

c. $x = \tan\theta$ and $y = \cot\theta$

d. $x = r\cos\theta$ where $r > 1$

By definition of the unit circle ($r = 1$), the horizontal coordinate is $x = \cos\theta$ and the vertical coordinate is $y = \sin\theta$.

2 Find the exact coordinates on the unit circle corresponding to $\theta = \frac{5\pi}{6}$.

a. $\left(-\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$ ✓

b. $\left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$

c. $\left(\frac{\sqrt{3}}{2}, -\frac{1}{2}\right)$

d. $\left(-\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$

In Quadrant II: $x = \cos(5\pi/6) = -\frac{\sqrt{3}}{2}$, $y = \sin(5\pi/6) = \frac{1}{2}$.

3 Find the exact coordinates on the unit circle for $\theta = \frac{4\pi}{3}$.

a. $\left(-\frac{1}{2}, -\frac{\sqrt{3}}{2}\right)$ ✓

b. $\left(-\frac{\sqrt{3}}{2}, -\frac{1}{2}\right)$

c. $\left(\frac{1}{2}, -\frac{\sqrt{3}}{2}\right)$

d. $\left(-\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right)$

In Quadrant III: $x = \cos(4\pi/3) = -1/2$, $y = \sin(4\pi/3) = -\sqrt{3}/2$.

Section 2: Reference Angles and Signs by Quadrant

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4 Find the reference angle θ' for $\theta = 225^\circ$.

a. 45° ✓

b. 35°

c. 135°

d. 55°

In Quadrant III: $\theta' = \theta - 180^\circ = 225^\circ - 180^\circ = 45^\circ$.

5 In which quadrants is $\tan\theta$ positive?

a. **Quadrants I and III** ✓

b. Quadrants I and II

c. Quadrants I and IV

d. Quadrants II and IV

$\tan\theta = y/x$. It is positive where x and y have the same sign: Quadrant I (both positive) and Quadrant III (both negative).

6 If $\cos\theta < 0$ and $\sin\theta > 0$, in which quadrant does the terminal ray of θ lie?

a. **Quadrant II** ✓

b. Quadrant I

c. Quadrant III

d. Quadrant IV

$x < 0$ and $y > 0$ defines Quadrant II.

Section 3: Exact Values of the Six Trig Functions

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7 Find the exact value of $\tan\left(\frac{2\pi}{3}\right)$.

a. $-\sqrt{3}$ ✓

b. $\sqrt{3}$

c. $-\frac{\sqrt{3}}{3}$

d. -1

Reference angle is $\pi/3$. In Quadrant II, tangent is negative: $-\tan(\pi/3) = -\sqrt{3}$.

8 Find the exact value of $\csc\left(\frac{7\pi}{4}\right)$.

a. $-\sqrt{2}$ ✓

b. $\sqrt{2}$

c. $-\frac{\sqrt{2}}{2}$

d. -2

$\sin(7\pi/4) = -\frac{\sqrt{2}}{2}$. Reciprocal gives $\csc(7\pi/4) = -\frac{2}{\sqrt{2}} = -\sqrt{2}$.

9 Evaluate $\cos(3\pi) - \sin\left(\frac{3\pi}{2}\right)$.

a. 0 ✓

b. -2

c. 2

d. 1

$\cos(3\pi) = \cos(\pi) = -1$. $\sin(3\pi/2) = -1$. Result $= -1 - (-1) = 0$.

Section 4: Quadrantal Angles and Negative Angles

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10 Which of the following values is undefined?

a. $\tan(90^\circ)$ ✓

b. $\sin(90^\circ)$

c. $\cos(90^\circ)$

d. $\cot(45^\circ)$

$\tan(90^\circ) = \frac{\sin(90^\circ)}{\cos(90^\circ)} = \frac{1}{0}$, which is undefined.

11 Evaluate $\cos(-\pi/3)$.

a. $1/2$ ✓

b. $-1/2$

c. $\sqrt{3}/2$

d. $-\sqrt{3}/2$

Cosine is an even function: $\cos(-\theta) = \cos\theta$. Thus $\cos(-\pi/3) = \cos(\pi/3) = 1/2$.

12 Evaluate $\sin(-\pi/4)$.

a. $-\frac{\sqrt{2}}{2}$ ✓

b. $\frac{\sqrt{2}}{2}$

c. -1

d. 0

Sine is an odd function: $\sin(-\theta) = -\sin\theta$. Thus $\sin(-\pi/4) = -\frac{\sqrt{2}}{2}$.

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